

# Lecture 12

## The Normal Distribution, z-scores and Transformations

# What you should know: kinds of study

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Experimental study

**Units are assigned to treatments at random: the study creates the groups.**

Observational study

**Variables are measured on each unit and nothing is assigned.**

Survey

**Selected units answer questions, to describe a population.**

# What you should know: sampling designs

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Simple random sample (SRS), and systematic sample

**SRS: every sample of  $n$  is equally likely. Systematic: random start, then every  $k$ th unit.**

Stratified random sample, and cluster sample

**Stratified: an SRS from every stratum. Cluster: randomly chosen whole clusters.**

Convenience sample, and voluntary response sample

**The easiest to reach, or self-selected. Both are biased at any sample size.**

# What you should know: experimental designs

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Completely randomized design

**Every unit is assigned to a treatment completely at random.**

Randomized block design

**Random assignment is done separately within blocks of similar units.**

Matched pairs design

**Blocks of two similar units, or one subject who receives both treatments.**

# What you should know: terminology

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The language of an experiment

**Units, factor, levels, treatments, control group, placebo, blinding.**

Association is not causation

**A lurking variable can create an association with no causal link.**

The two things randomness buys you

**Random sampling lets you generalize. Random assignment lets you claim cause.**

# Data: female student heights

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A survey of 379 college students recorded each student's height and gender; 262 are women.

**X = the height of a female student, in inches**

| HEIGHT | GENDER |
|--------|--------|
| 56     | Female |
| 57     | Female |
| 58     | Female |
| 60     | Female |
| 60     | Female |
| ...    | ...    |

n = 262 women (the file is sorted from shortest to tallest)

# Data: Central Park in January

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The National Weather Service has recorded the mean temperature in Central Park, New York, every month since 1869.

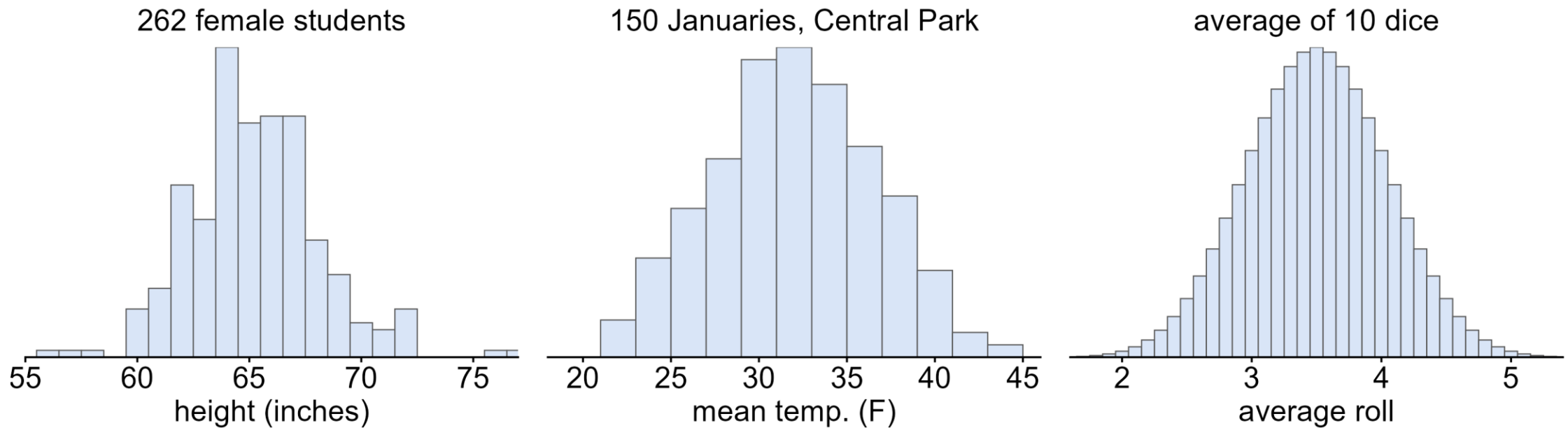
**X = the mean January temperature in Central Park in a given year, in °F**

| YEAR | JAN  | FEB  | MAR  | ... | DEC  | ANNUAL |
|------|------|------|------|-----|------|--------|
| 1869 | 35.1 | 34.5 | 34.8 | ... | 34.7 | 51.4   |
| 1870 | 37.5 | 31.3 | 34.1 | ... | 34.1 | 53.6   |
| 1871 | 28.3 | 30.2 | 44.2 | ... | 29.2 | 51.1   |
| 1872 | 28.8 | 29.9 | 30.5 | ... | 26.7 | 51     |
| 1873 | 28.6 | 29.5 | 35.7 | ... | 36.5 | 51     |
| ...  | ...  | ...  | ...  | ... | ...  | ...    |

n = 150 years, 1869 to 2018

# What do these have in common?

**What do these three distributions have in common?**

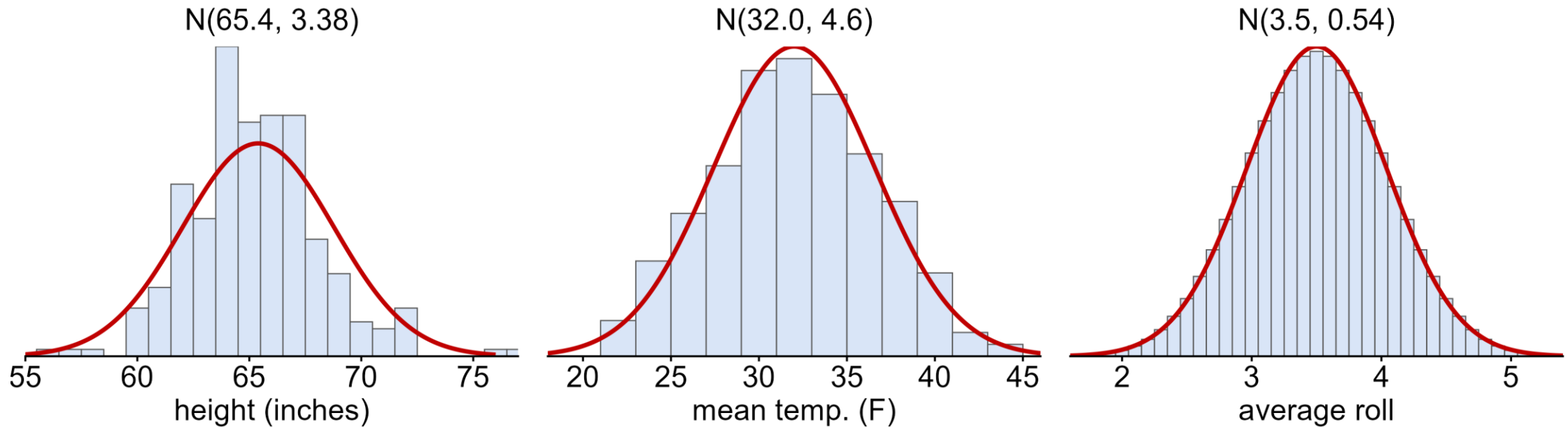


**Each is single-peaked, symmetric and bell shaped**

# The normal distribution

## normal distribution

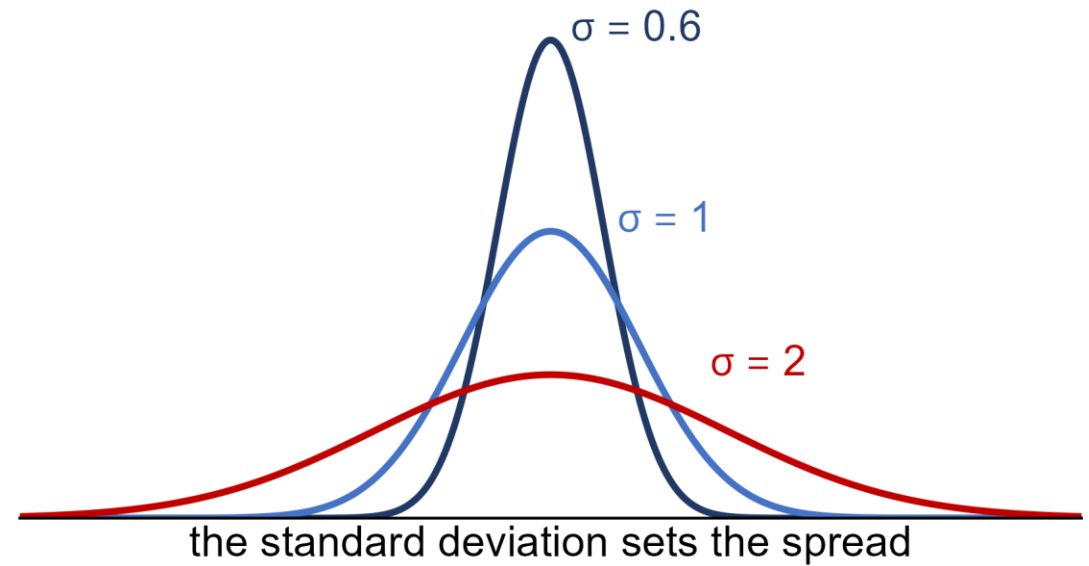
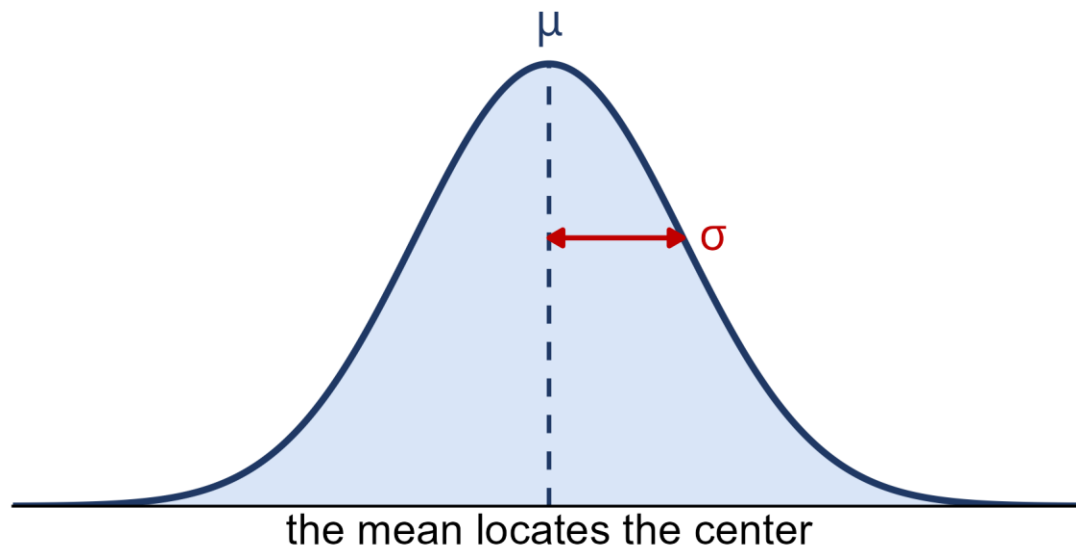
a symmetric, bell-shaped curve with its peak at the mean



# Two parameters: $\mu$ and $\sigma$

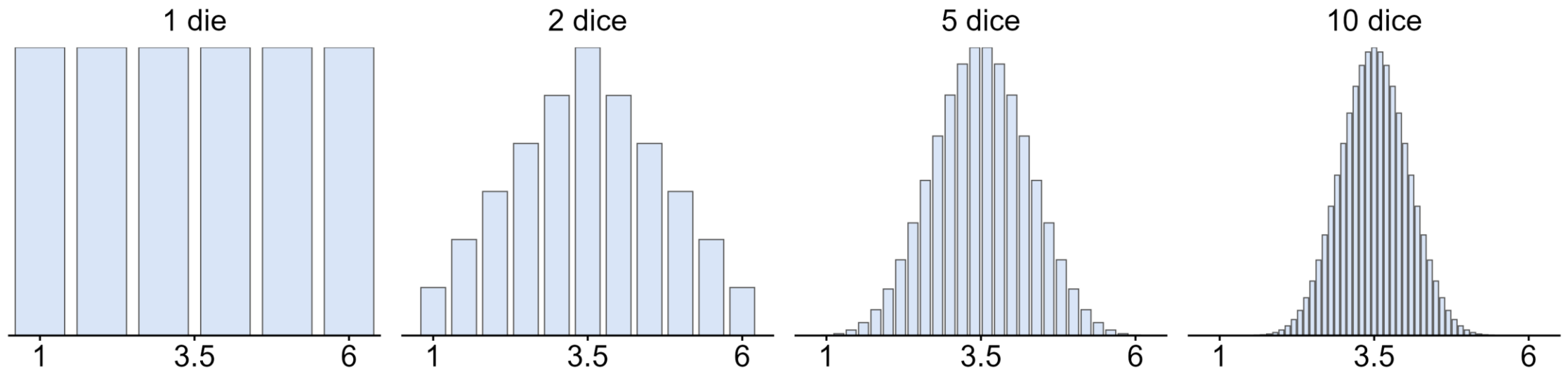
$$X \sim N(\mu, \sigma)$$

$\mu$  locates the center;  $\sigma$  sets the spread



# Why is the normal so important?

**Why does one curve matter so much?**



**Many measurements are approximately normal**

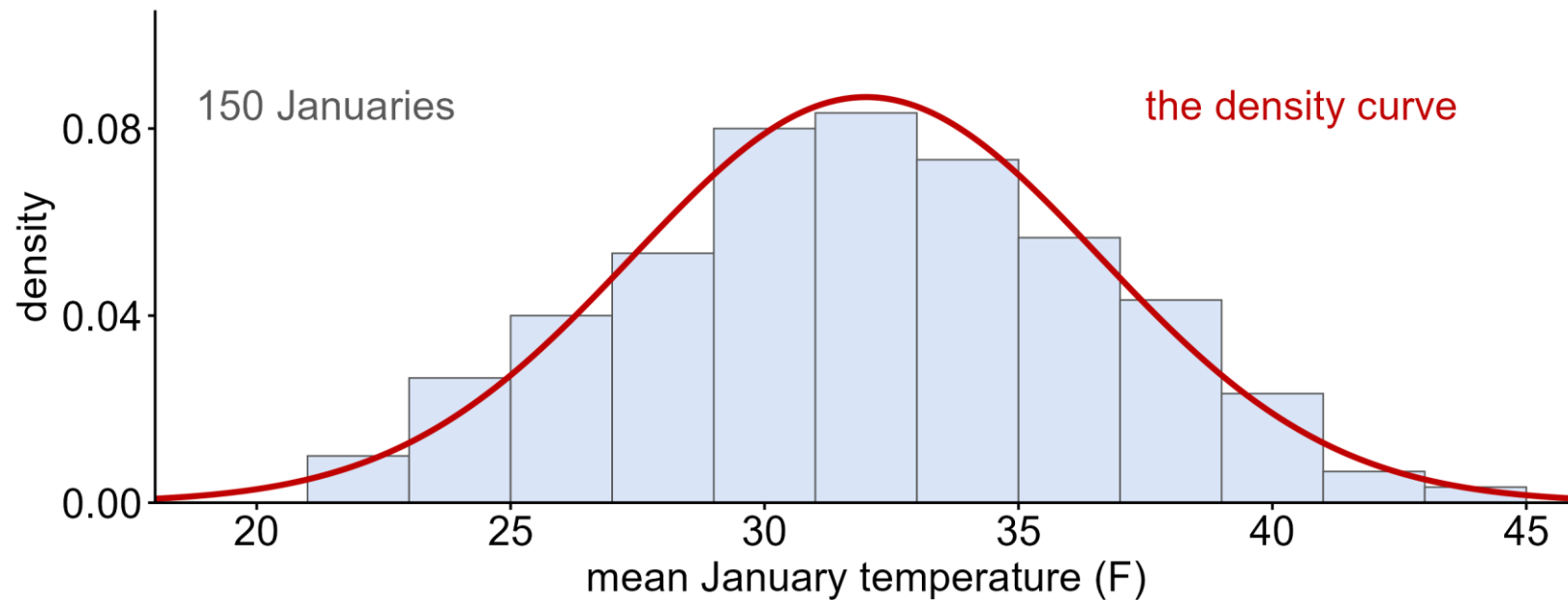
**Averages of random values become normal, whatever the data look like**

**Most methods in this course are built on it**

# Density curves

## density curve

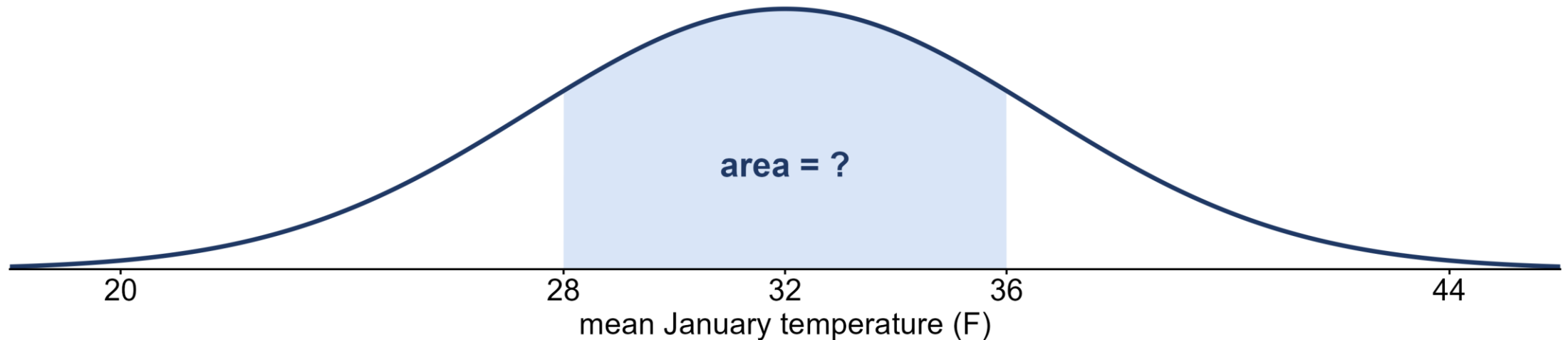
a smooth curve that describes a distribution; the total area under it is 1



# Area under the curve is proportion

Central Park, 150 Januaries:  $\bar{x} = 32.0^\circ\text{F}$   $s = 4.6^\circ\text{F}$

**What fraction of Januaries averaged between  $28^\circ\text{F}$  and  $36^\circ\text{F}$ ?**



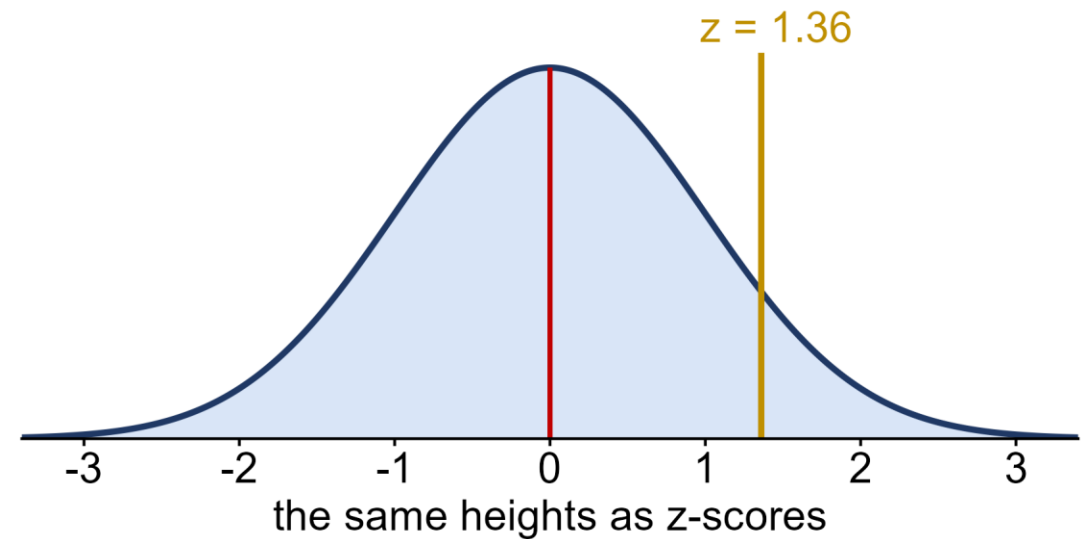
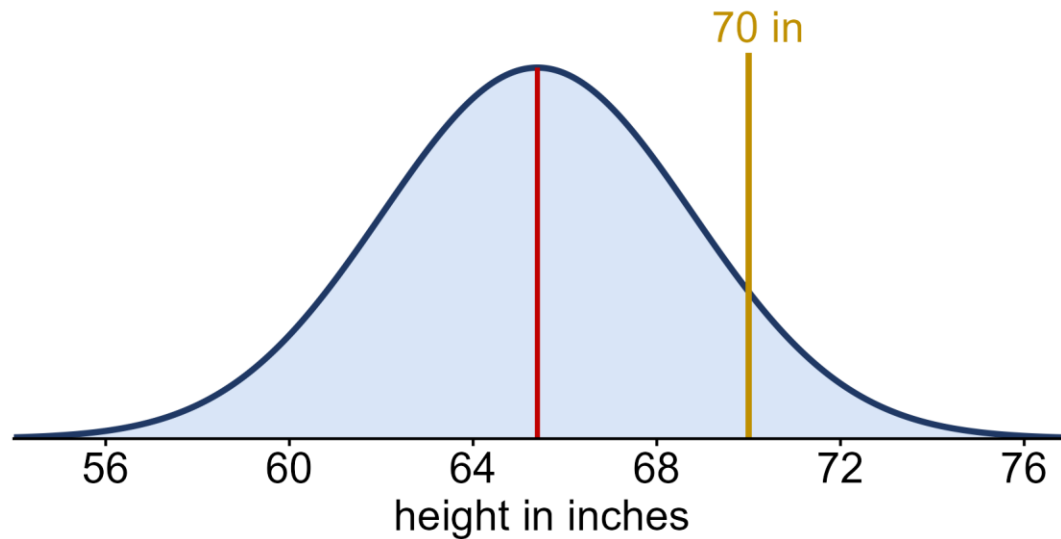
**Area under the curve: 0.62      Actual: 92 of 150 = 0.61**

**The height of the curve is density: where values crowd. The area is the proportion.**

# The standard normal distribution

**standard normal distribution**

the normal distribution with mean 0 and standard deviation 1, written  $Z \sim N(0, 1)$



# Data: breakfast cereals

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The twenty US breakfast cereals from Week 3, with their sodium and sugar content.

**Y = the sodium content of a breakfast cereal, in mg**

| Cereal              | Sodium | Sugar | Type |
|---------------------|--------|-------|------|
| Frosted Mini Wheats | 0      | 11    | A    |
| Raisin Bran         | 340    | 18    | A    |
| All Bran            | 70     | 5     | A    |
| Apple Jacks         | 140    | 14    | C    |
| Captain Crunch      | 200    | 12    | C    |
| ...                 | ...    | ...   | ...  |

n = 20 cereals

# Which observation is more extreme?

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$X$  = an exam score (points) and  $Y$  = the sodium in a breakfast cereal (mg); each sample has  $n = 20$ .

**Sample 1 ( $X$ ):  $\bar{x} = 74$ ,  $s = 8$ , observation  $x = 90$**

**Sample 2 ( $Y$ ):  $\bar{y} = 167$ ,  $s = 77$ , observation  $y = 290$**

**Which observation is more extreme?**

**$x = 90$ :  $16 / 8 = 2.0$  standard deviations above its mean**

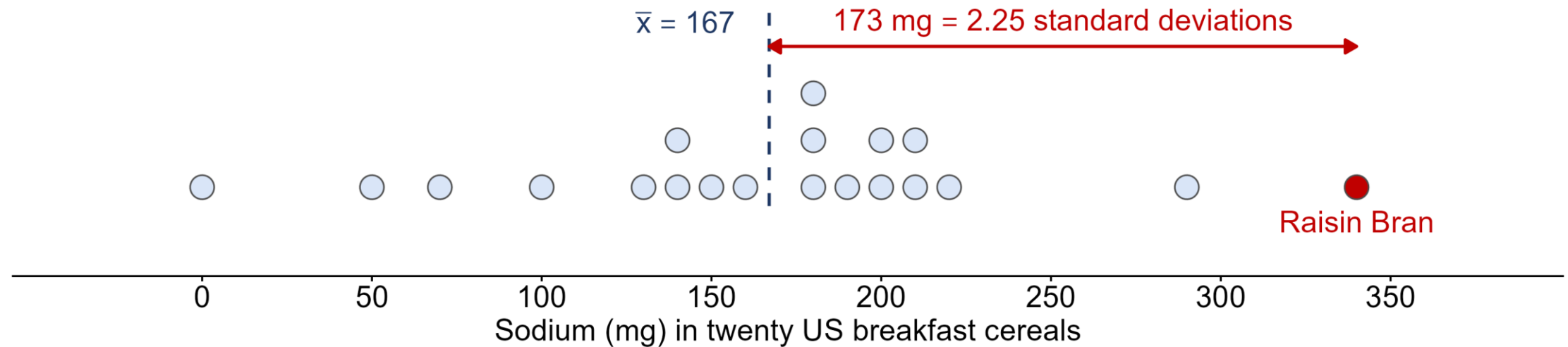
**$y = 290$ :  $123 / 77 \approx 1.6$  standard deviations above its mean**

**$x = 90$  is more extreme**

# Measuring a deviation in standard deviations

How unusual is the sodium in Raisin Bran, among the 20 cereals from Week 3?

$x = 340 \text{ mg}$      $\bar{x} = 167 \text{ mg}$      $s = 77 \text{ mg}$



$$z = (x - \bar{x}) / s = 173 / 77 = 2.25 \rightarrow \text{the z-score}$$

# Example: Jerome's points

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Jerome averages 16 points a game, and one night he scores 10 (textbook Try It 6.2).

$$X \sim N(16, 4) \quad x = 10$$

$x = 10$  is \_\_\_\_ standard deviations to the \_\_\_\_ (right or left) of the mean \_\_\_\_.

$$z = (10 - 16) / 4 = -1.5$$

1.5 standard deviations to the left of the mean, 16

# Data: EU carbon emissions

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The World Bank's carbon dioxide emissions per person for the 28 member states of the European Union.

**X = the carbon dioxide emissions per person in an EU country, in metric tons**

| Country Name | Country Code | CO2 |
|--------------|--------------|-----|
| Austria      | AUT          | 8   |
| Belgium      | BEL          | 10  |
| Bulgaria     | BGR          | 6   |
| Croatia      | HRV          | 4.7 |
| Cyprus       | CYP          | 7   |
| ...          | ...          | ... |

n = 28 countries

# Example: carbon emissions across the EU

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Carbon emissions per person in the 28 EU member states (World Bank).

$\bar{x} = 7.9 \text{ t}$       $s = 3.6 \text{ t}$      Luxembourg = 21.4 t

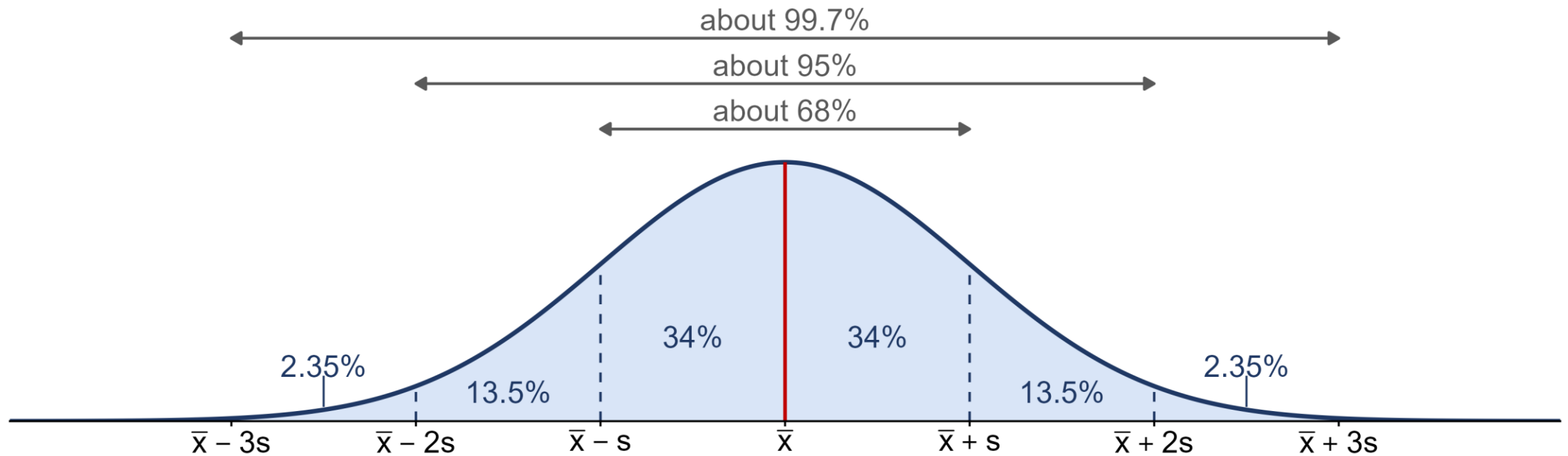
- (a) How many standard deviations above the mean is Luxembourg?
- (b) What emission level sits 2 standard deviations above the mean?

- (a)  $z = 3.75$
- (b) 15.1 t, and only Luxembourg is above it

# The empirical rule

## empirical rule

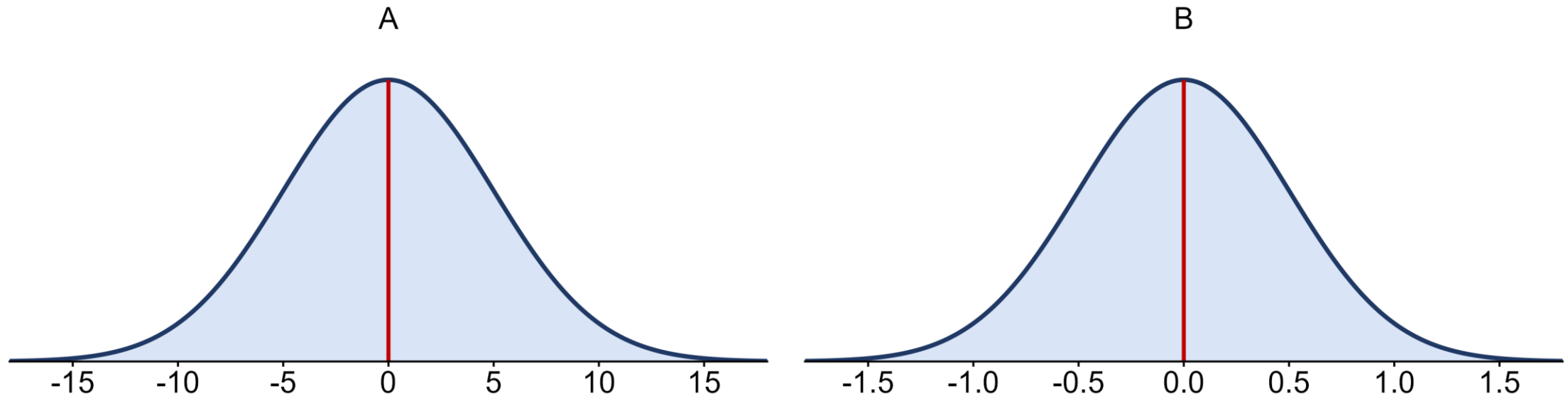
about 68% of values fall within 1 standard deviation of the mean, 95% within 2, and 99.7% within 3



# What is the standard deviation?

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Estimate  $\sigma$  for curve A and for curve B.



**A:  $\sigma \approx 5$**

**B:  $\sigma \approx 0.5$**

# Example: back to that exam

Exam scores in the class are roughly bell shaped.

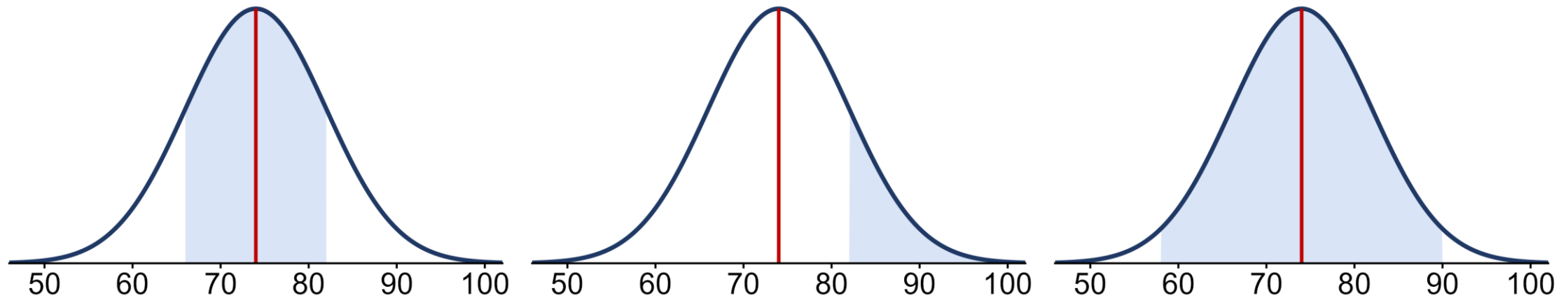
$\bar{x} = 74$  points       $s = 8$  points

**What percentage of scores falls in each shaded region?**

(a) 66 to 82 points

(b) 82 points and above

(c) 58 to 90 points



**(a) about 68%**

**(b) about 16%**

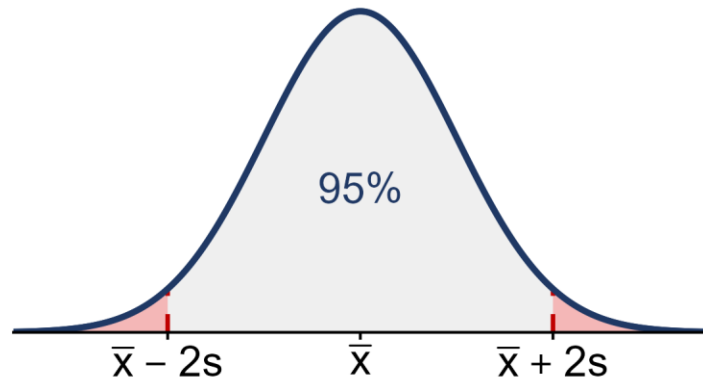
**(c) about 95%**

# Outliers: the two-standard-deviation rule

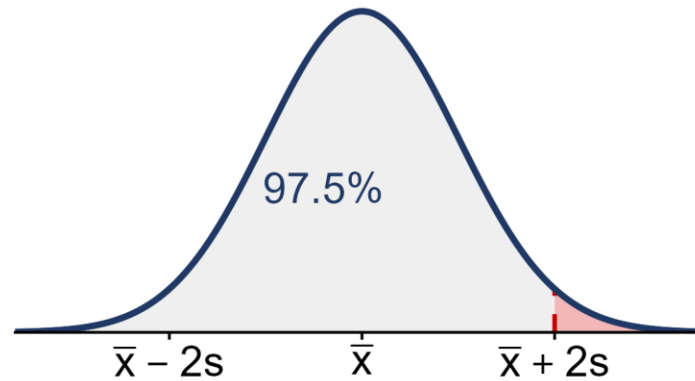
## outlier

a value at least 2 standard deviations from the mean,  $|z| \geq 2$  (about 5% of bell-shaped data)

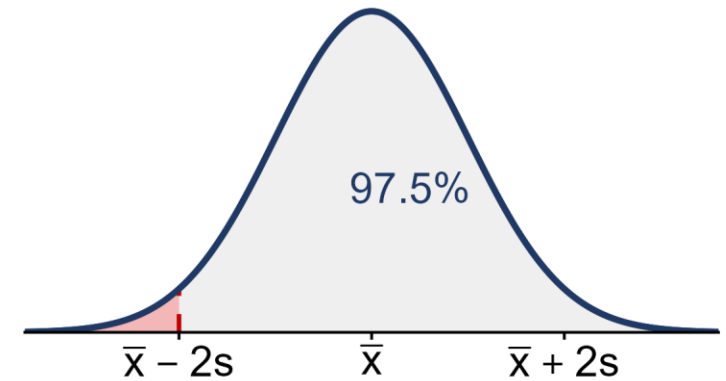
both tails:  $2.5\% + 2.5\% = 5\%$



upper tail only: 2.5%

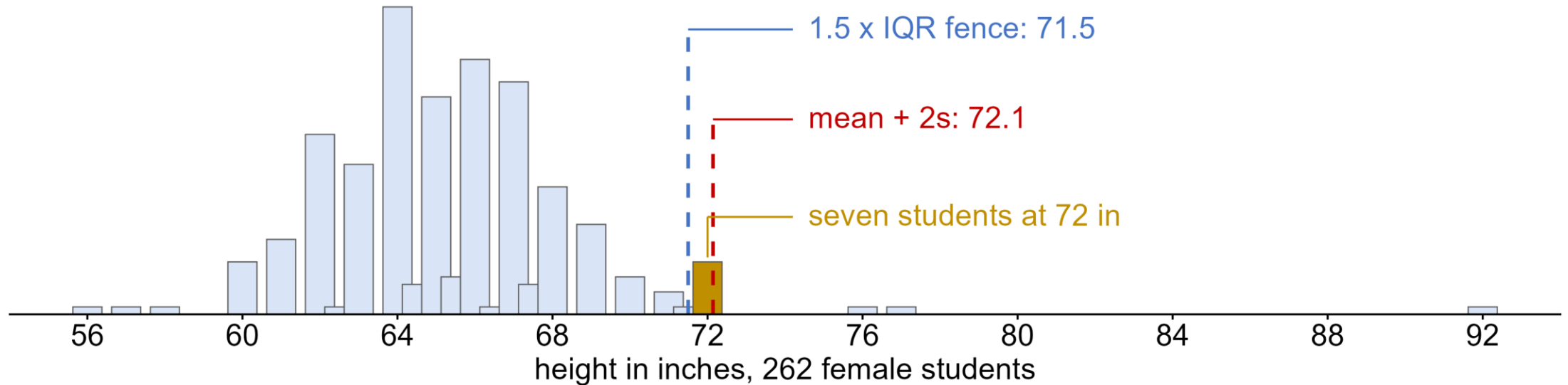


lower tail only: 2.5%



# When the two outlier rules disagree

Do the  $1.5 \times \text{IQR}$  rule and the  $\pm 2s$  rule flag the same students?



**No: the IQR rule flags 13 students, the  $\pm 2s$  rule only 6**

**Why: the 92-inch value inflates  $s$  and pushes the  $2s$  fence out; the quartiles barely move**

# Computing quantiles from a normal distribution

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Heights of 262 female students are roughly bell shaped.

$\bar{x} = 65.4$  in       $s = 3.38$  in      one student is 70 in

- (a) What is her z-score?
- (b) What percentage of the women are shorter than she is?

- (a)  $z = 1.36$
- (b) The empirical rule cannot say: 1.36 is not a whole number of standard deviations

# How it used to be done: the Z-table

the standard normal table

each entry is the area to the LEFT of  $z$

| $z$ | .00    | .01    | .02    | .03    | .04    | .05    | .06           | .07    | .08    | .09    |
|-----|--------|--------|--------|--------|--------|--------|---------------|--------|--------|--------|
| 1.0 | 0.8413 | 0.8438 | 0.8461 | 0.8485 | 0.8508 | 0.8531 | 0.8554        | 0.8577 | 0.8599 | 0.8621 |
| 1.1 | 0.8643 | 0.8665 | 0.8686 | 0.8708 | 0.8729 | 0.8749 | 0.8770        | 0.8790 | 0.8810 | 0.8830 |
| 1.2 | 0.8849 | 0.8869 | 0.8888 | 0.8907 | 0.8925 | 0.8944 | 0.8962        | 0.8980 | 0.8997 | 0.9015 |
| 1.3 | 0.9032 | 0.9049 | 0.9066 | 0.9082 | 0.9099 | 0.9115 | <b>0.9131</b> | 0.9147 | 0.9162 | 0.9177 |
| 1.4 | 0.9192 | 0.9207 | 0.9222 | 0.9236 | 0.9251 | 0.9265 | 0.9279        | 0.9292 | 0.9306 | 0.9319 |
| 1.5 | 0.9332 | 0.9345 | 0.9357 | 0.9370 | 0.9382 | 0.9394 | 0.9406        | 0.9418 | 0.9429 | 0.9441 |
| 1.6 | 0.9452 | 0.9463 | 0.9474 | 0.9484 | 0.9495 | 0.9505 | 0.9515        | 0.9525 | 0.9535 | 0.9545 |

$z = 1.36$ : row 1.3, column .06 → 0.9131, so about 91% are shorter

# Today: the computer does the lookup

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From a value to an area: pnorm

**pnorm(1.36) = 0.9131**

From an area to a value: qnorm

**qnorm(0.90) = 1.28**

No R in front of you?

**The normal calculator at [stat-tools.jmks-homelab.dev](http://stat-tools.jmks-homelab.dev) does both.**

# A note about transformations of variables

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Changing units

**We often need to change the units of a variable: °F to °C, feet to meters, dollars to euros.**

Linear transformations: adding, subtracting, multiplying, dividing

**$y = ax + b$  (a scales, b shifts). The z-score is a linear transformation. They preserve the shape of the distribution.**

Nonlinear transformations: squaring, taking roots, logarithms, exponentiation

**They do NOT preserve the shape of the distribution.**

# What a linear transformation does to the summaries

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The center moves with the whole transformation

**$\bar{y} = a\bar{x} + b$ , and the same for the median.**

The spread ignores the shift

**$s$  and IQR are multiplied by  $|a|$ ;  $b$  has no effect.**

The shape does not change at all

**Skewness, gaps and outliers all survive unchanged.**

# Example: inches to centimeters

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The 262 heights are converted from inches to centimeters:  $y = 2.54x$ .

$\bar{x} = 65.4$  in       $s = 3.38$  in      the 70-inch student = 177.8 cm

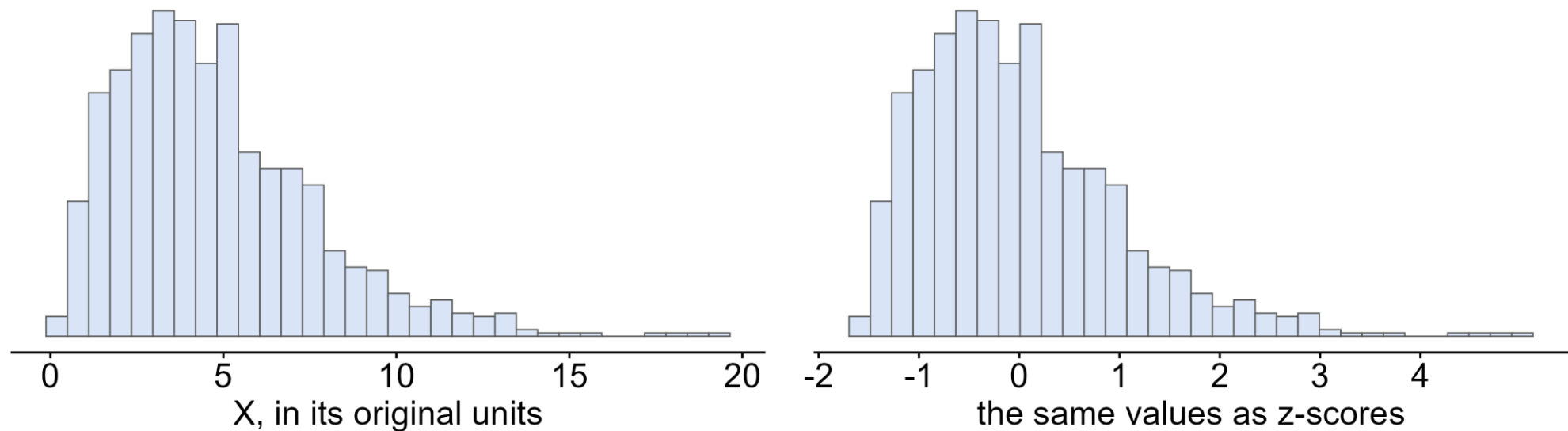
- (a) What are the mean and standard deviation in centimeters?
- (b) What is her z-score in centimeters?

(a)  $\bar{y} = 166.1$  cm,  $s = 8.59$  cm

(b)  $z = 1.36$ , unchanged

# Does a z-score make data normal?

**Does converting skewed data to z-scores make it normal?**

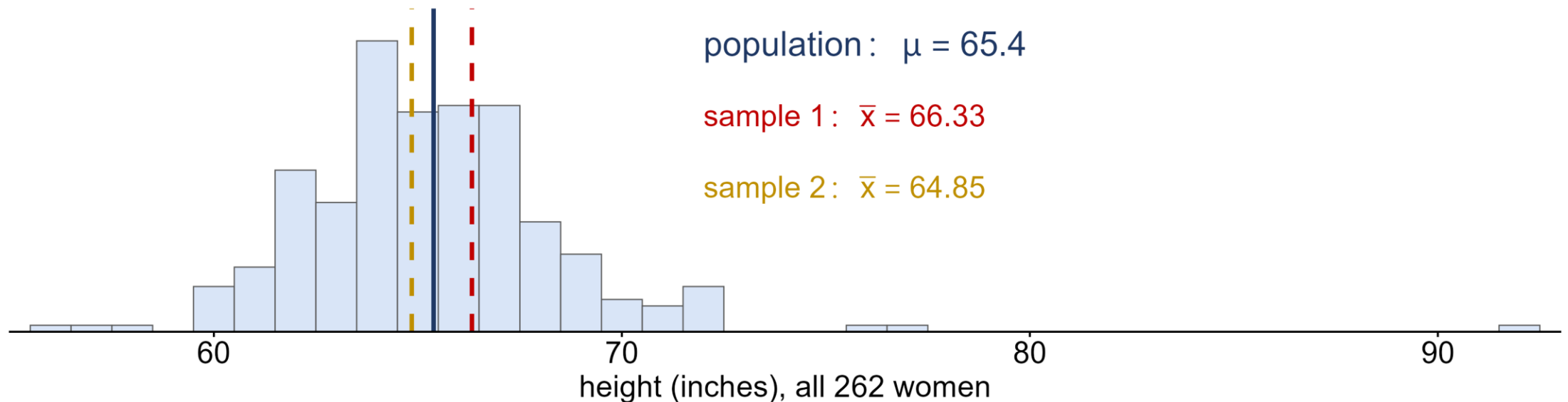


**No: standardizing changes the axis, not the shape**

# Estimating a mean from a sample

Treat the 262 women as the population, and estimate  $\mu$  from a random sample of  $n = 20$ .

**Why do the two samples miss  $\mu$ , and by different amounts?**

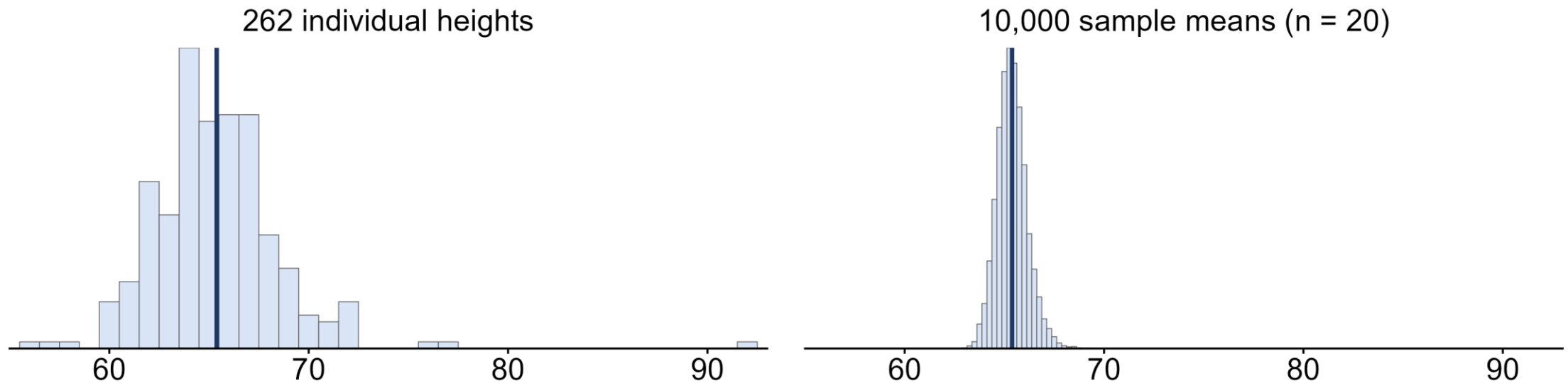


**Each sample holds different students, so each  $\bar{x}$  misses by a different amount: sampling error**

# Sampling distributions

## sampling distribution

the distribution of a statistic, such as  $\bar{x}$ , over all possible samples of the same size  $n$



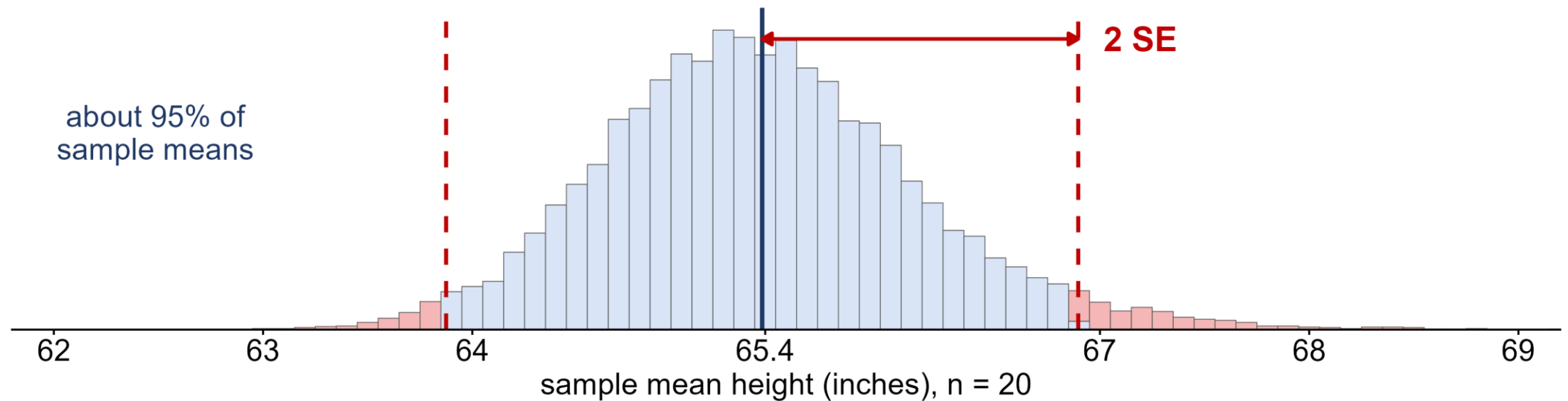
**You've seen these before:**

**$SE = s / \sqrt{n}$  (Week 3) is its spread; the Rec Center histograms (Week 4) were sampling distributions**

# Margin of error

## margin of error

how far we expect an estimate to fall from the parameter: the largest distance to an estimate that is not an outlier,  $2 \times \text{SE}$



# Example: the margin of error for $\bar{x}$

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Random samples of  $n = 20$  from the 262 women.

$\sigma = 3.38$  in       $n = 20$

What are the standard error and the margin of error of  $\bar{x}$ ?

$$SE = \sigma / \sqrt{n} = 3.38 / \sqrt{20} \approx 0.76 \text{ in}$$

Margin of error =  $2 \times 0.76 \approx 1.5$  in: about 95% of samples land within  $65.4 \pm 1.5$  in

# Statistical significance

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Someone hands you a random sample of 20 students and says they are from our 262 women.

$\bar{x} = 70.33$  in       $\mu = 65.4$  in       $SE = 0.76$  in

**Is this result statistically significant?**

**$z = (70.33 - 65.4) / 0.76 \approx 6.5$ : far outside the margin of error, so yes**  
**Statistically significant: too large to be ordinary sampling variation (they were men)**

# What you should know: the normal distribution

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The two parameters

$x \sim N(\mu, \sigma)$ :  $\mu$  locates the center,  $\sigma$  sets the spread.

The empirical rule

68%, 95% and 99.7% within 1, 2 and 3 standard deviations (bell-shaped only).

The z-score

$z = (x - \bar{x}) / s$  counts standard deviations from the mean;  $|z| \geq 2$  flags an outlier.